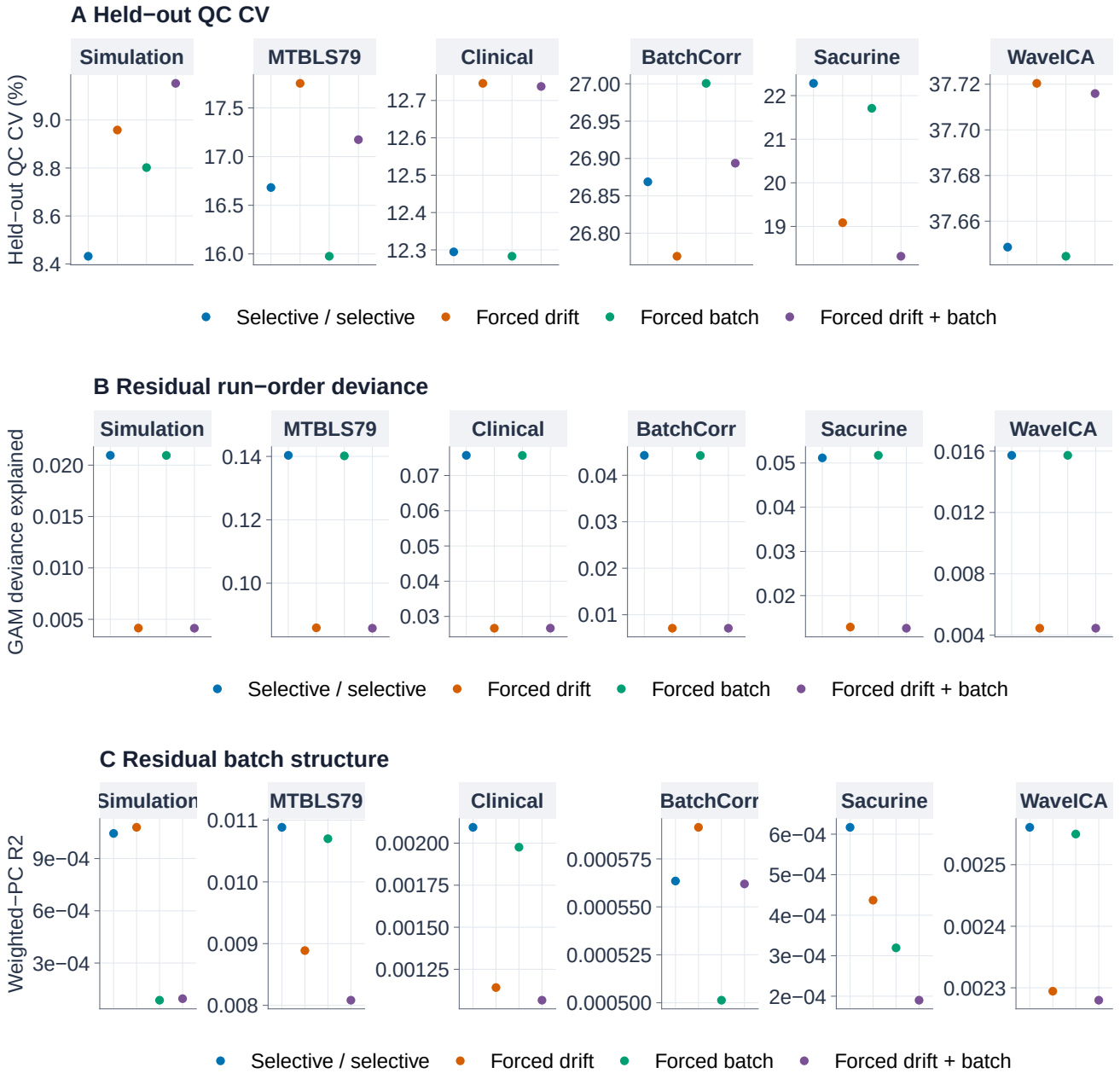
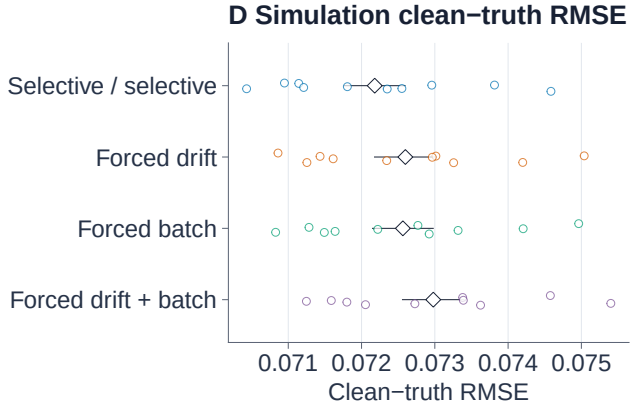
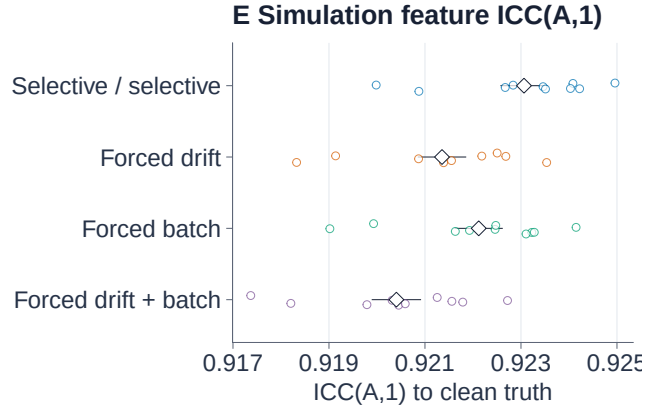
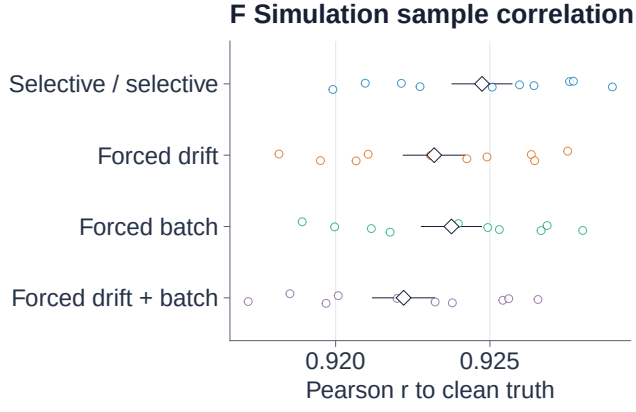

A Held−out QC CV
Simulation
MTBLS79
Clinical
BatchCorr
Sacurine
WaveICA
Held−out QC CV (%)
9.0
8.8
8.6
8.4
17.5
17.0
16.5
16.0
12.7
12.6
12.5
12.4
12.3
27.00
26.95
26.90
26.85
26.80
22
21
20
19
37.72
37.70
37.68
37.66
Selective / selective
Forced drift
Forced batch
Forced drift + batch
B Residual run−order deviance
GAM deviance explained
0.020
0.015
0.010
0.005
0.14
0.12
0.10
0.07
0.06
0.05
0.04
0.03
0.04
0.03
0.02
0.01
0.05
0.04
0.03
0.02
0.016
0.012
0.008
0.004
Selective / selective
Forced drift
Forced batch
Forced drift + batch
C Residual batch structure
Simulation
MTBLS79
Clinical
BatchCorr
Sacurine
WaveICA
Weighted−PC R2
9e−04
6e−04
3e−04
0.011
0.010
0.009
0.008
0.00200
0.00175
0.00150
0.00125
0.000575
0.000550
0.000525
0.000500
6e−04
5e−04
4e−04
3e−04
2e−04
0.0025
0.0024
0.0023
Selective / selective
Forced drift
Forced batch
Forced drift + batch


D Simulation clean−truth RMSE
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.071
0.072
0.073
0.074
0.075
Clean−truth RMSE


E Simulation feature ICC(A,1)
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.917
0.919
0.921
0.923
0.925
ICC(A,1) to clean truth


F Simulation sample correlation
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.920
0.925
Pearson r to clean truth